

PES UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
Software Engineering Lab (UE24CS242) — Lab 4: VibeCoding Report

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Semester & Branch:	4th Semester, B.Tech CSE
Assigned Repository:	https://github.com/SETAPESU26/31_galaga
Lab Repository:	https://github.com/Mohith-1502/SE-LAB-PES1UG24CS580
Lab Deliverables Folder:	Lab-4

1. Executive Summary & Lab Objectives

This laboratory exercise focuses on applying modern Vibe Coding methodologies to repair, enhance, and ship a feature-rich arcade game (Galaga-lite in Pygame). Through precise, context-rich prompting under strict budget constraints (< 4 prompts), the objective is to resolve mathematical curve defects and systematically implement core gameplay mechanics.

2. Vibe Coding Prompts & Implementation History

Task 1: Bug Fix: Cubic Bézier Curve Trajectory Overshoot

Technical Diagnosis: In game.py, bezier() computes positions along a cubic curve. The standard mathematical formula is $B(t) = (1-t)^3P_0 + 3(1-t)^2tP_1 + 3(1-t)t^2P_2 + t^3P_3$. In the provided starter code, the third term was erroneously written as $3*u*t^2[p2[0]]$ with exponent 1 instead of 2 on t. This broke polynomial weight normalization and caused enemies to bulge and overshoot mid-flight.

Vibe Coding Prompt:

In game.py, examine bezier(p0, p1, p2, p3, t). Enemy entry and dive paths overshoot and bulge unnaturally mid-flight even though endpoints match. Compare the four weighted terms against standard cubic Bézier formula:

$B(t) = (1-t)^3P_0 + 3*(1-t)^2*tP_1 + 3*(1-t)*t^2P_2 + t^3P_3$

Fix the missing exponent on t in the third term for both x and y coordinates.

Result & Outcome: Corrected $3 * u * t$ to $3 * u * t * t$ for both x and y coordinates in bezier(). Smooth, graceful flight paths restored.

Git Commit: 9981dca fix: correct cubic bezier curve calculation in bezier()

Task 2: Feature: Dynamic Wave-Based Enemy Recolor (enemy_tint)

Technical Diagnosis: enemy_tint(kind) was an empty function called per enemy in Game.draw(). By tying enemy coloration to wave progression, each wave presents a fresh, distinctive cyber-arcade visual theme.

Vibe Coding Prompt:

Implement enemy_tint(kind) in game.py. The function receives kind ('boss', 'red', or 'blue') and should return an (r, g, b) color tuple or None. Recolor enemies dynamically based on current wave number so that subsequent waves look visually distinct and challenging.

Result & Outcome: Configured WAVE_PALETTES with custom neon/cyber palettes indexed by current_wave. Enemies dynamically adopt new palettes per wave.

Git Commit: 43c6005 feat: implement enemy_tint() for dynamic wave-based enemy recoloring

Task 3: Feature: Stage Announcement Banner (on_wave_start)

Technical Diagnosis: on_wave_start(wave) is called once per wave when enemies are generated. To provide classic arcade polish, a timed banner alert notifies the player of stage transitions.

Vibe Coding Prompt:
Implement on_wave_start(wave) in game.py. It is called when a wave spawns. When invoked, track the wave number and show a retro arcade 'STAGE - READY' announcement banner for 2 seconds in the center of the screen with a golden border and timed fadeout.

Result & Outcome: Implemented on_wave_start to track current_wave and set a 2.2-second wave_banner. Integrated timer countdown in Game.update and centered bordered banner rendering in Game.draw.

Git Commit: 26ae4e6 feat: implement on_wave_start() with stage announcement banner

Task 4: Feature: Player Energy Shield & Visual Feedback (shield_charges)

Technical Diagnosis: shield_charges(wave) sets player shield capacity each wave. Granting 1 charge initially and +1 every 3 waves balances survival difficulty, complemented by a cyan elliptical aura and HUD tracking.

Vibe Coding Prompt:
Implement shield_charges(wave) in game.py to grant 1 shield charge at wave 1 plus 1 additional charge every 3 waves. In Game.draw(), display current shield charges on HUD and draw an active cyan energy shield ring around the player ship when shield charges > 0.

Result & Outcome: Returns 1 + (wave - 1) // 3. Renders Shield counter in HUD and adds an animated cyan energy ellipse surrounding active player ships.

Git Commit: cf4085c feat: implement shield_charges() and visual energy shield aura

3. Deliverables Verification Checklist

Deliverable Item	Artifact Location	Status
Before Gameplay Video (15s)	Lab-4/before.mp4	Verified
After Gameplay Video (15s)	Lab-4/after.mp4	Verified
Updated Source Code	game.py & Lab-4/code/game.py	Verified
Chat History Word Document	Lab-4/PES1UG24CS580_Lab4_VibeCoding_ChatHistory.docx	Verified
Chat History PDF Document	Lab-4/PES1UG24CS580_Lab4_VibeCoding_ChatHistory.pdf	Verified